

Algebraic Topology: a beginner's course

This lecture:

**AlgTop3: Two-dimensional
surfaces: the sphere**

copyright 2010 N J Wildberger

Lecture 3: Two-dimensional surfaces: the sphere.



$$= [0, 0, -1] + t(x, y, 1) \quad t = \frac{z}{x^2 + y^2 + 1}$$

$$= [tx, ty, t-1] = Q$$

choose t so that Q on sphere:

$$x^2 + (t-1)^2 = 1 \quad \therefore Q = \left[\frac{2x}{1+x^2+y^2}, \frac{2y}{1+x^2+y^2}, \frac{1-x^2-y^2}{1+x^2+y^2} \right]$$

$$(t-1) - 2t = 0$$

$$(t-1) - 2t = 0$$

Compare: $e(h) = \left(\frac{1-h^2}{1+h^2}, \frac{2h}{1+h^2} \right)$

Algebraic Topology: a beginner's course

next in the series:

**AlgTop4:
More on the sphere**